

Yichao Brightfield-to-Fluorescence Pix2Pix Result Report

OrganoidAgent

May 11, 2026

Summary

This report documents the corrected same-time brightfield-to-fluorescence (B2F) experiment for the Yichao projected organoid instance dataset. The goal is to establish whether the paired brightfield crop contains enough morphological information to reconstruct or classify fluorescence before using the model for future-expression prediction.

The successful run used the older pix2pix-style U-Net family that previously worked well in `/home/lachlan/ProjectsLFS/OrganoidAgent/pixel2pixel_fluorescent/fluorescent/cyclegan_modules.py`, adapted to the projected instance-pair database. Training used original 384 pixel crops, no AMP, balanced sampling, and resumable early stopping. The run stopped at epoch 78 after validation convergence was detected; the best validation checkpoint was epoch 10.

Data and Model

The training data came from the projected instance-pair manifest:

$$\mathcal{D} = \{(B_i, F_i, M_i)\}_{i=1}^N,$$

where B_i is the brightfield crop, F_i is the corresponding fluorescence crop, and M_i is the organoid mask. The split sizes were 6691 training instances, 1601 validation instances, and 1088 held-out test instances.

The model was a pix2pix-style U-Net with InstanceNorm and ELU blocks. For each brightfield image, the network predicts fluorescence logits:

$$\hat{F}_i = \sigma(G_\theta(B_i)).$$

The objective combined weighted fluorescence reconstruction, pixel BCE, structure, edge, and scalar auxiliary terms. Bright fluorescence pixels received higher weight so the sparse signal was not overwhelmed by dark background.

Training and Early Stopping

The early-stop score was reconstruction-first:

$$s = -\mathcal{L}_{val} + 0.05 \text{F1}_{signal} + 0.01 r_{peak}.$$

Training stopped when this score failed to improve by at least 0.001 for 10 validation checks after epoch 40. On resume, the trainer scanned the existing metrics and detected that the model had already converged.

B2F training convergence and early stopping

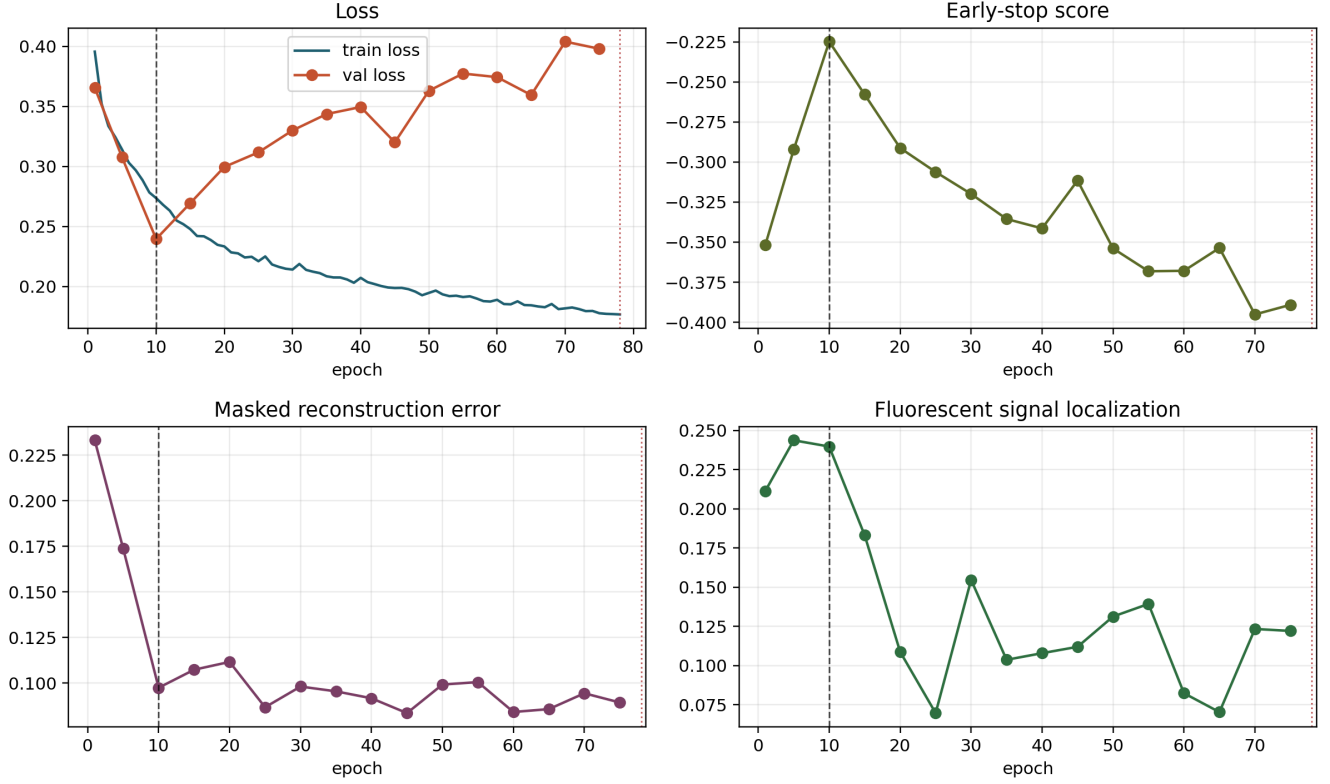


Figure 1: Training convergence, validation loss, early-stop score, masked MAE, and fluorescent signal F1. The best checkpoint occurred early, while later epochs continued improving training loss but degraded validation score.

Held-Out Quantification

Table 1: Main held-out test metrics from the best checkpoint.

Quantity	Value
Best epoch	10
Stop epoch	78
Test signal F1	0.387
Test signal precision	0.266
Test signal recall	0.710
Test expression AUROC	0.879
Test masked MAE	0.096
Test PSNR	18.03 dB

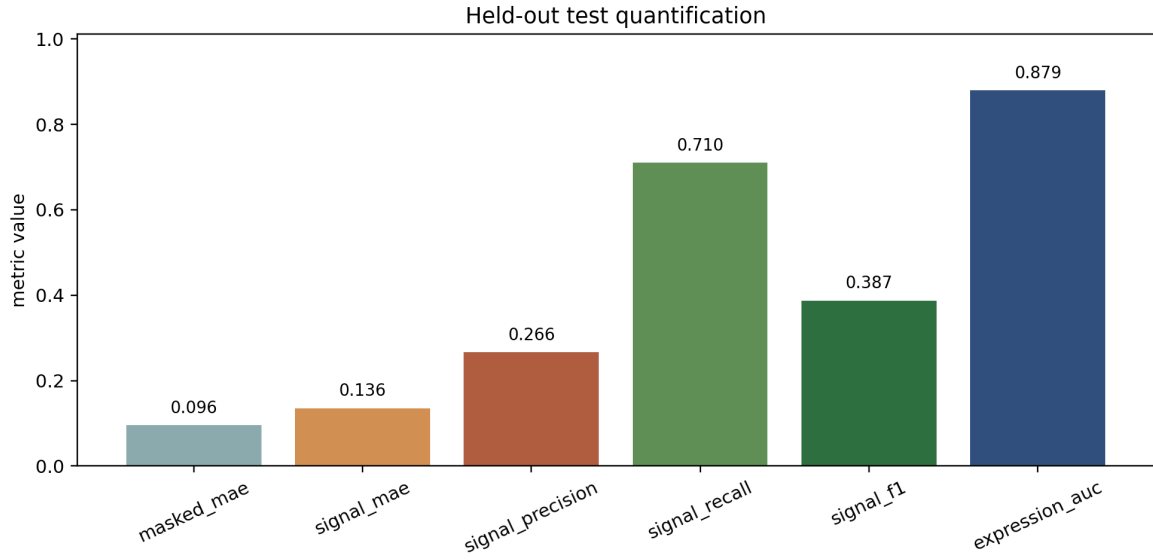


Figure 2: Aggregate test-set metrics. The model has high recall and moderate precision, consistent with a useful but over-inclusive fluorescence predictor.

Table 2: Per-dataset held-out test summary. True and predicted signal values are fractions of masked organoid pixels above the fluorescence threshold.

Dataset	n	True signal	Pred signal	Signal F1	Masked MAE
Data-Yichao-1	1	0.000	0.165	0.003	0.145
Data-Yichao-2	33	0.154	0.124	0.178	0.122
Data-Yichao-3	345	0.053	0.290	0.130	0.117
Data-Yichao-5	180	0.002	0.103	0.007	0.099
Data-Yichao-6	86	0.018	0.087	0.039	0.078
Data-Yichao-7	170	0.024	0.053	0.064	0.060
Data-Yichao-9	273	0.370	0.745	0.501	0.098

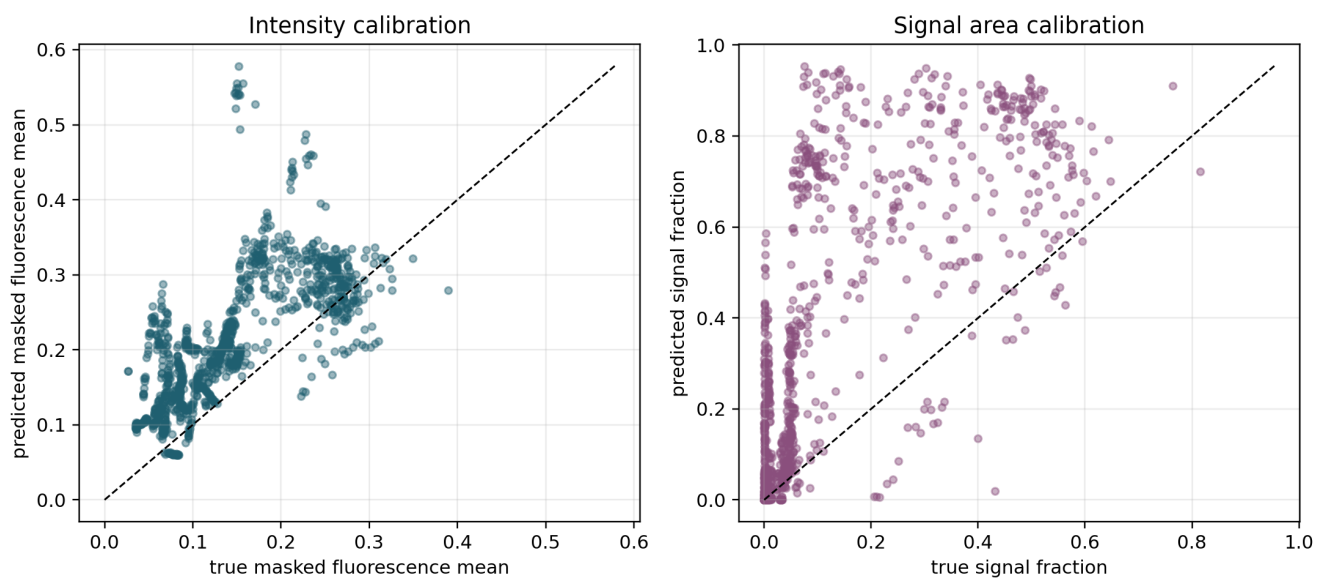


Figure 3: Per-instance calibration on the held-out test set. Left: masked mean fluorescence. Right: fluorescent signal area. The masked mean correlation was 0.692, but signal area tends to be over-predicted for weak or negative examples.

Prediction Visualizations

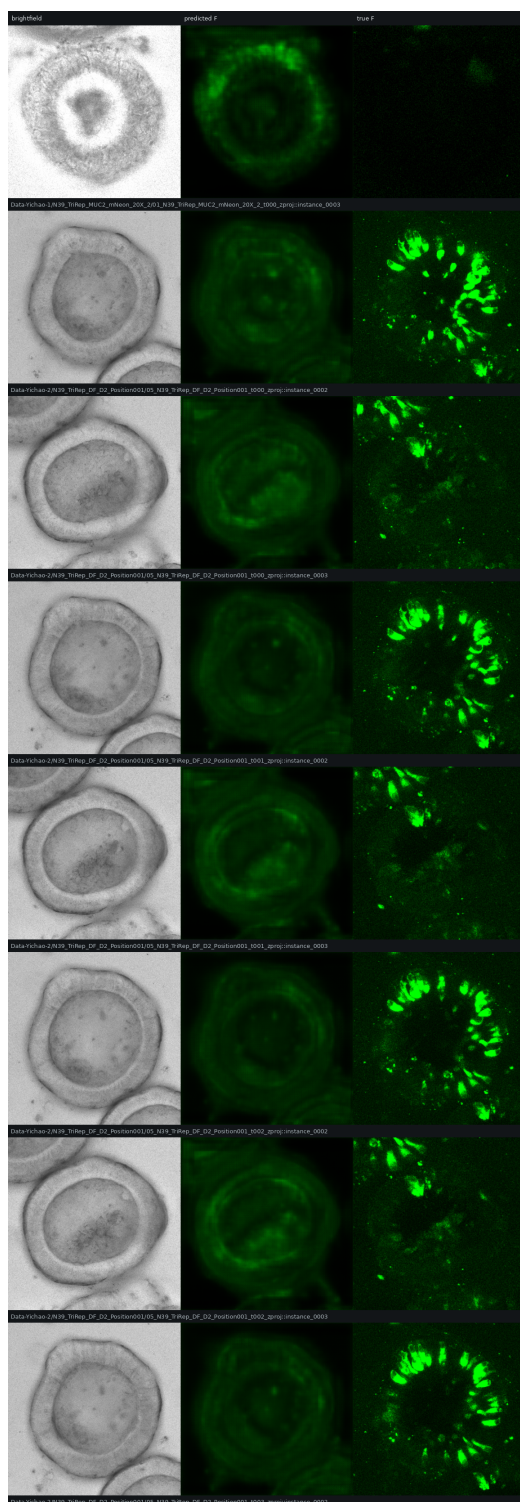


Figure 4: Original held-out test panel saved by the trainer: brightfield input, predicted fluorescence, and true fluorescence.

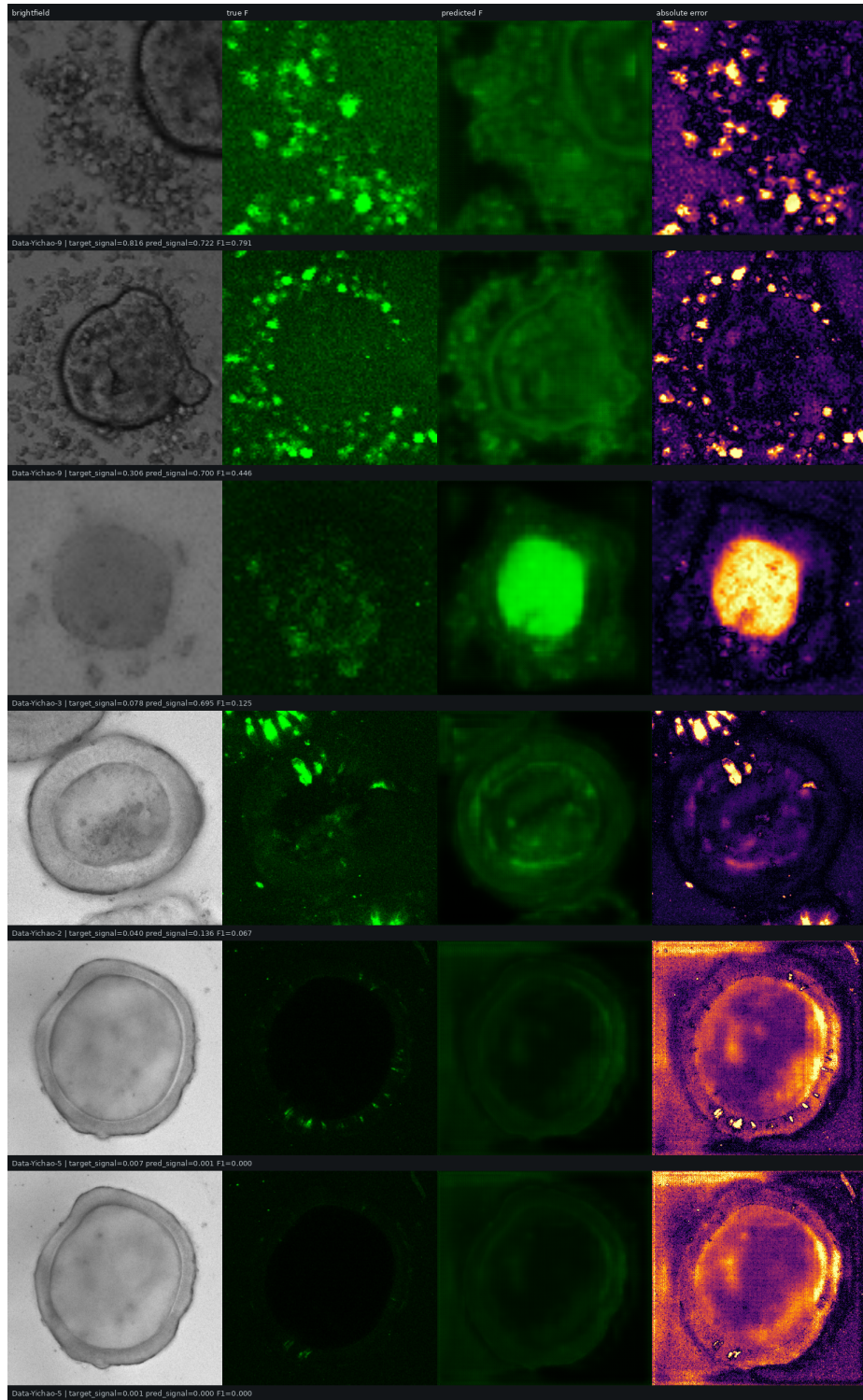


Figure 5: Selected held-out examples with brightfield input, true fluorescence, predicted fluorescence, and absolute error. The model often captures broad positive regions, but can over-predict signal in low-expression cases.

Explanation

The explanation figure uses input-gradient saliency. For each selected held-out crop, the objective was the mean predicted fluorescence inside the organoid mask:

$$a(B) = \frac{1}{|M|} \sum_{x \in M} \hat{F}(x), \quad S(x) = \left| \frac{\partial a}{\partial B(x)} \right|.$$

This is a model-sensitivity map, not a causal biological proof. It indicates which brightfield pixels the trained B2F model most relied on when increasing predicted fluorescence.

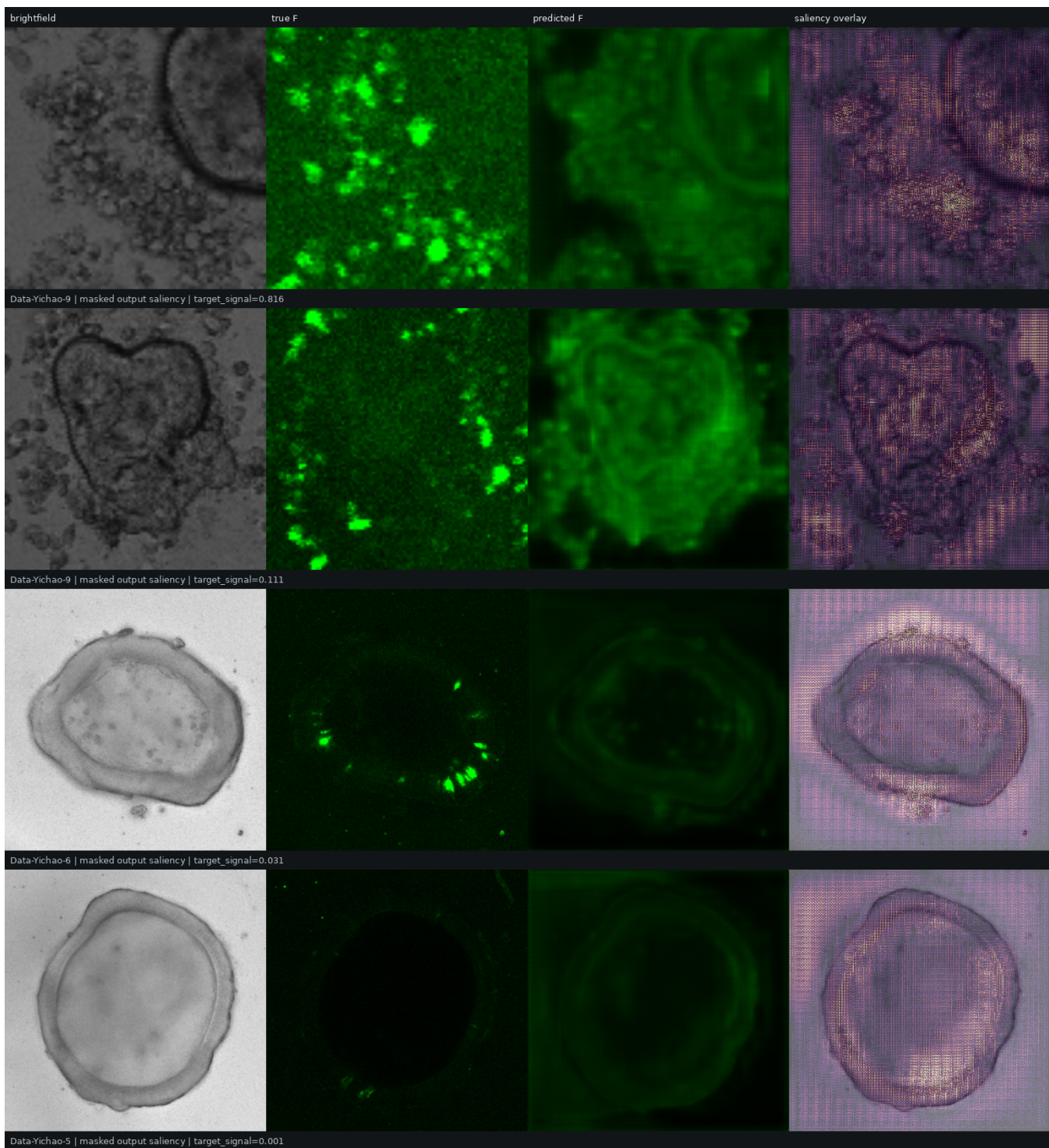


Figure 6: Gradient saliency overlays. The highlighted regions tend to follow organoid texture and boundary structure, supporting the interpretation that the model is using morphology rather than a constant intensity prior.

Interpretation

The corrected B2F experiment is now technically valid: optimizer steps occurred, validation was monitored, and early stopping selected the best checkpoint. The held-out test AUROC for expression classification

was 0.879, indicating that brightfield morphology contains information associated with fluorescence-positive status.

The reconstruction is not yet perfect. Signal recall was 0.710, but precision was 0.266, so the model tends to predict too much fluorescent area. The per-dataset table shows that performance is strongest on Data-Yichao-9, while low-expression datasets expose over-prediction. This matters for the next phase: future-expression prediction should not use raw predicted pixel area alone. It should combine calibrated B2F outputs with morphology, time, position, and expression onset analysis.

Artifacts

The artifact generator is:

```
/home/lachlan/ProjectsLFS/OrganoidAgent/differentiation_prediction/yichao_future_expression/  
make_b2f_result_artifacts.py
```

The finished run is:

```
/home/lachlan/ProjectsLFS/OrganoidAgent/analysis-outputs/yichao_future_expression/stage1_b2f_  
pix2pix_384_v1_noamp_long
```

The generated per-instance test quantification table is:

```
/home/lachlan/ProjectsLFS/OrganoidAgent/publication/yichao_b2f_pix2pix_results_report/tables/  
test_per_instance_quantification.csv
```